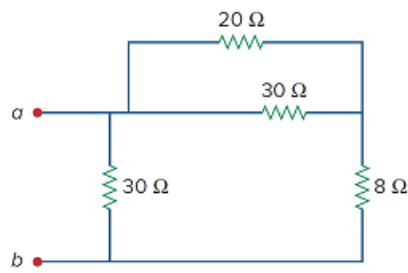


Problem

For the circuits in Fig. 2.108, obtain the equivalent resistance at terminals $a-b$.

Figure 2.108:



Step-by-step solution

Step 1 of 2

Refer to Figure 2.108 in the text book.

The resistors $20\ \Omega$ and $30\ \Omega$ are connected in parallel.

$$\begin{aligned} 20\ \Omega \parallel 30\ \Omega &= \frac{20 \times 30}{20 + 30} \\ &= \frac{600}{50} \\ &= 12\ \Omega \end{aligned}$$

Step 2 of 2

The parallel combination is connected in series with $8\ \Omega$ resistor.

$$12\ \Omega + 8\ \Omega = 20\ \Omega$$

The equivalent resistance obtained and $30\ \Omega$ are connected in parallel.

$$\begin{aligned} 20\ \Omega \parallel 30\ \Omega &= \frac{20 \times 30}{20 + 30} \\ &= \frac{600}{50} \\ &= 12\ \Omega \end{aligned}$$

Therefore, the equivalent resistance at terminals $a-b$ is $\boxed{12\ \Omega}$.